AI Decision Journal

Project Title: Hostel Maintenance Ticket Sorter

1. How I Used AI

I used Gemini AI to read hostel maintenance complaints and understand the problems. It helps to find four details: room number, type of problem, urgency and a short summary. Gemini AI wrote the first draft of the code. I rewrote and tested each function myself, fixed the errors, and changed the model name when the old one was retired.

2. AI Decision Records

No.	Decision	Why did I choose it?	How did I check it?
1	Used Gemini AI to read complaints.	It saves time when handling many complaints.	Python checks the information given by AI.
2	Used AI to find room numbers.	It helps to know where the problem is.	Python checks whether the room number is correct and present in the complaint.
3	Used AI to find the type of problem.	It helps to send the complaint to the correct team.	Python checks the issue type and assigns the team.
4	Used AI to find the urgency of complaints.	Urgent problems need quick attention.	Python checks for dangerous words and increases urgency when needed.
5	Added safety checks in Python.	AI may sometimes miss dangerous problems.	Python checks words like spark, shock, overflowing and burning smell.
6	Added duplicate checking.	The same complaint should not create many tickets.	Python checks the room number and issue with existing tickets.
7	Added due-date rules.	Every complaint needs a deadline.	Urgent tickets get the same-day deadline, while normal tickets get two days.
8	Added a warden check.	Important decisions should not be made by AI alone.	The warden checks every urgent ticket and manages the work.

3. Problems I Faced

While testing, the free Gemini service returned 503 errors because it was busy, and later the free daily limit of 20 requests was reached. I did not use a fake or local classifier. I tested each Python function separately, and I tested one real AI call, which returned valid JSON. The full run of 24 complaints could not finish in one day because of the limit.

4. What I Learned

- AI can help us complete tasks faster.
- We should check the information given by AI.
- Safety checks are important.
- Duplicate complaints should be handled properly.
- API request limits should be considered.
- Human checking is important for urgent complaints.

5. Conclusion

Through this project, I learned how to use AI with Python to manage hostel complaints. I understood that AI can make work easier, but we should also use proper checks and human supervision to make the system work correctly.